# Quantum 2 HW4

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*Technion*

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# 1 The Optical Theorem

## 1.1 Why $S$ is unitary

In an elastic scattering the kinetic energy of the incident particle is conserved. Therefore applying the operator S to a state does not change its magnitude in the energy space, therefore it is unitary.

## 1.2 Show VS

We know that

$$H_0 |\psi_i\rangle = E |\psi_i\rangle \tag{1}$$
$$H |\psi_o\rangle = E |\psi_o\rangle \tag{2}$$

inserting the full form of the hamiltonian

$$(H_0 + V) |\psi_o\rangle = E |\psi_o\rangle \tag{3}$$

subtracting a zero on the right hand side

$$(H_0 + V) |\psi_o\rangle = E |\psi_o\rangle - 0 \tag{4}$$
$$(H_0 + V) |\psi_o\rangle = E |\psi_o\rangle - (E - H_0) |\psi_i\rangle \tag{5}$$
$$(H_0 + V) S |\psi_i\rangle = ES |\psi_i\rangle - (E - H_0) |\psi_i\rangle \tag{6}$$
$$VS |\psi_i\rangle = ES |\psi_i\rangle - (E - H_0) |\psi_i\rangle - H_0 S |\psi_i\rangle \tag{7}$$
$$VS = ES - E + H_0 - H_0 S \tag{8}$$
$$VS = (E - H_0)(S - \mathbb{1}) \tag{9}$$

in the limit $v \to 0$ S is the identity, since both sides of the equation must equal 0 always.

## 1.3 Resolvant

If we apply $G_0$ to both sides

$$G_0 VS = G_0 (E - H_0)(S - \mathbb{1}) \implies G_0 VS = S - \mathbb{1} \implies S = \mathbb{1} + G_0 VS \tag{10}$$

### 1.4 Projection

We'll apply the above equation to the state $|k\rangle$

$$S|k\rangle = \mathbb{1}|k\rangle + G_0 V S|k\rangle \tag{11}$$
$$|\psi_o\rangle = |k\rangle + G_0 V|\psi_o\rangle \tag{12}$$

### 1.5 Collision Operator

directly

$$S = \mathbb{1} + G_0 V S \implies S = \mathbb{1} + G_0 T \tag{13}$$

### 1.6 Applying to the State

applying state $|k\rangle$

$$S|k\rangle = \mathbb{1}|k\rangle + G_0 T|k\rangle \tag{14}$$
$$|\psi_o\rangle = |k\rangle + G_0 T|k\rangle \tag{15}$$

This is another representation of the Lippman Schwinger equation.

### 1.7 Projection Again

the form we found for $\psi_o(r)$ is

$$\psi_o(r) = \frac{1}{(2\pi)^{3/2}}\left(e^{i\boldsymbol{k}\cdot\boldsymbol{r}} + \frac{e^{ikr}}{r}f(k,k')\right) \tag{16}$$

where

$$f(k,k') = -\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3\langle k'|V|\psi_o\rangle \tag{17}$$

transforming using our known operators

$$f(k,k') = -\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3\langle k'|VS|k\rangle \tag{18}$$
$$= -\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3\langle k'|T|k\rangle \tag{19}$$

### 1.8 Showing Things

directly

$$V|\psi_o\rangle = VS|k\rangle = T|k\rangle \tag{20}$$

using LS we found earlier

$$V|\psi_o\rangle = V|k\rangle + VG_0T|k\rangle \tag{21}$$
$$T|k\rangle = V|k\rangle + VG_0T|k\rangle \tag{22}$$
$$T = V + VG_0T \tag{23}$$

### 1.9 Showing More Things

$$\begin{aligned}
\Im\left(\langle k|T|k\rangle\right) &= \Im\left(\langle k|VS|k\rangle\right) && (24)\\
&= \Im\left(\langle k|V|\psi_o\rangle\right) && (25)\\
&= \Im\left(\langle \psi_o|V|\psi_o\rangle - \langle \psi_o|VG_0V|\psi_o\rangle\right) && (26)\\
&= \Im\left(\langle \psi_o|V|\psi_o\rangle\right) - \Im\left(\langle \psi_o|VG_0V|\psi_o\rangle\right) && (27)\\
&= \Im\left(\langle \psi_o|V|\psi_o\rangle\right) - \Im\left(\langle \psi_o|V\left(\Re G_0 + i\Im G_0\right)V|\psi_o\rangle\right) && (28)\\
&= -\Im\left(\langle \psi_o|V\Re G_0V|\psi_o\rangle\right) + \Im\left(i\langle \psi_o|V\Im G_0V|\psi_o\rangle\right) && (29)\\
&= -\Im\left(i\langle \psi_o|V\Im G_0V|\psi_o\rangle\right) && (30)\\
&= -\Im\left(i\langle \psi_o|V\Im\left(\frac{1}{E-H_0+o\varepsilon}\right)V|\psi_o\rangle\right) && (31)\\
&= -\pi\Im\left(i\langle \psi_o|V\delta\left(E-H_0\right)V|\psi_o\rangle\right) && (32)\\
&= -\pi\langle \psi_o|V\delta\left(E-H_0\right)V|\psi_o\rangle && (33)
\end{aligned}$$

### 1.10 Conclude

$$\begin{aligned}
-\pi\langle \psi_o|V\delta\left(E-H_0\right)V|\psi_o\rangle &= -\pi\langle k|S^\dagger V\delta\left(E-H_0\right)VS|k\rangle && (34)\\
&= -\pi\langle k|T^\dagger\delta\left(E-H_0\right)T|k\rangle && (35)
\end{aligned}$$

inserting an identity

$$-\pi\langle k|T^\dagger\delta\left(E-H_0\right)T|k\rangle = -\pi\int \mathrm{d}k'\,\langle k|T^\dagger\delta\left(E-H_0\right)|k'\rangle\langle k'|T|k\rangle \quad (36)$$

next we notice that since the momentum is $P=\hbar k$ the hamiltonian is $H_0=\frac{\hbar^2k^2}{2m}$

$$\begin{aligned}
-\pi\int \mathrm{d}k'\,\langle k|T^\dagger\delta\left(E-H_0\right)|k'\rangle\langle k'|T|k\rangle &= -\pi\int \mathrm{d}k'\delta\left(E-\frac{\hbar^2k'^2}{2m}\right)\langle k|T^\dagger|k'\rangle\langle k'|T|k\rangle && (37)\\
&= -\pi\int \mathrm{d}k'\delta\left(E-\frac{\hbar^2k'^2}{2m}\right)|\langle k'|T|k\rangle|^2 && (38)
\end{aligned}$$

moving integration parameter to the energy we notice $\frac{\mathrm{d}E}{\mathrm{d}k}=\frac{\hbar^2k}{m}$ therefore $\frac{\mathrm{d}k}{\mathrm{d}E}=\frac{m}{\hbar^2k}$ and remembering we're now integrating over a sphere in energy phase space

$$\begin{aligned}
-\pi\int \mathrm{d}k'\delta\left(E-\frac{\hbar^2k'^2}{2m}\right)|\langle k'|T|k\rangle|^2 &= -\pi\int \mathrm{d}E\,\mathrm{d}\Omega k^2\frac{m}{\hbar^2k}\delta\left(E-\frac{\hbar^2k'^2}{2m}\right)|\langle k'|T|k\rangle|^2 && (39)\\
&= -\frac{\pi m}{\hbar^2k}\int \mathrm{d}E\,\mathrm{d}\Omega k^2\delta\left(E-\frac{\hbar^2k'^2}{2m}\right)|\langle k'|T|k\rangle|^2 && (40)\\
&= -\frac{\pi mk}{\hbar^2}\int \mathrm{d}\Omega|\langle k'|T|k\rangle|^2 && (41)
\end{aligned}$$

## 1.11 Prove Optical Theorem

In subsection 7 we found

$$f(k,k') = -\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3 \langle k'|T|k\rangle \tag{42}$$

therefore

$$\begin{aligned}
\Im\left(f(\theta=0)\right) &= \Im\left(f(k,k)\right) && (43)\\
&= -\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3 \,\Im\left(\langle k|T|k\rangle\right) && (44)\\
&= -\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3 \left(-\frac{\pi mk}{\hbar^2}\int \mathrm{d}\Omega |\langle k'|T|k\rangle|^2\right) && (45)\\
&= \frac{k}{2\hbar^4}(2\pi)^3 \int \mathrm{d}\Omega |\langle k'|T|k\rangle|^2 && (46)\\
&= \frac{k}{2\hbar^4}(2\pi)^3 \left(-\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3\right)^{-2} \int \mathrm{d}\Omega |f(k,k)| && (47)\\
&= \frac{k}{4\pi}\int \mathrm{d}\Omega |f(k,k)| && (48)
\end{aligned}$$

using the definition for the differential cross section $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = |f(k,k')|^2$

$$\begin{aligned}
\Im\left(f(\theta=0)\right) &= \frac{k}{4\pi}\int \mathrm{d}\Omega \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} && (49)\\
&= \frac{k}{4\pi}\sigma && (50)
\end{aligned}$$

# 2 Quantization of the Field

1. Calculate the commutators $[v_x, H]$, $[v_y, H]$, $[v_z, H]$
2. Use Ehrenfest's theorem to calculate

$$C_1(t) = \langle S_z v_z \rangle \qquad C_2(t) = \langle S_x v_x + S_y v_y \rangle \qquad C_3(t) = \langle S_x v_y - S_y v_x \rangle \tag{51}$$

3. Write the time development of $\langle \boldsymbol{S} \cdot \boldsymbol{v} \rangle$ as a function of $C_1(0), C_2(0), C_3(0)$
4. a beam of electrons moving at velocity $\boldsymbol{v}$ is prepared at $t = 0$. The electrons in the beam have spin such that $C_1(0), C_2(0), C_3(0)$ are known. The electrons in the beam interact with the magnetic field $\boldsymbol{B}$ in the time frame $[0, T]$. At time $t = T$ a value corresponding to $\langle \boldsymbol{S} \cdot \boldsymbol{v} \rangle$. Conclude $a$ from the graph.
5. Compare the experiment result to theory.

## 2.1 Commutators

We'll represent both all the operators using the $\Pi$ operators as we did in the lecture. If we define

$$\Pi_i = \frac{p_i - aA_i}{\sqrt{\hbar e B}} \tag{52}$$

then

$$H = \frac{\hbar\omega_c}{2} \left( \Pi_x^2 + \Pi_y^2 + \Pi_z^2 - \sigma_z (1 + a) \right) \tag{53}$$

and

$$\boldsymbol{v} = \frac{\sqrt{\hbar e B}}{m} \boldsymbol{\Pi} \tag{54}$$

therefore

$$[v_x, H] = \frac{\sqrt{\hbar e B}}{m} \frac{\hbar\omega_c}{2} [\Pi_x, \Pi_y^2] = i\hbar\omega_c v_y \tag{55}$$

$$[v_y, H] = \frac{\sqrt{\hbar e B}}{m} \frac{\hbar\omega_c}{2} [\Pi_y, \Pi_x^2] \tag{56}$$

$$[v_z, H] = 0 \tag{57}$$

## 2.2 Ehrenfest

To prepare I'll calculate the commutators

$$[S_x, H] = -\frac{\hbar^2\omega_c}{4} (1 + a) [\sigma_x, \sigma_z] = i\hbar\omega_c (1 + a) S_y \tag{58}$$

$$[S_y, H] = -\frac{\hbar^2\omega_c}{4} (1 + a) [\sigma_y, \sigma_z] = -i\hbar\omega_c (1 + a) S_x \tag{59}$$

$$[S_z, H] \propto [\sigma_z, \sigma_z] = 0 \tag{60}$$

and thus

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}C_1 &= \frac{1}{i\hbar}\langle[S_z v_z, H]\rangle = 0 && (61)\\
\frac{\mathrm{d}}{\mathrm{d}t}C_2 &= \frac{1}{i\hbar}\langle[S_x v_x + S_y v_y, H]\rangle = 0 && (62)\\
&= \frac{1}{i\hbar}\langle i\hbar\omega_c\left(1+a\right)S_y v_x + i\hbar\omega_c v_y S_x - i\hbar\omega_c\left(1+a\right)S_x v_y - i\hbar\omega_c S_y v_x\rangle && (63)\\
&= \Omega\langle S_y v_x + S_x v_y\rangle && (64)\\
&= -\Omega C_3 && (65)\\
\frac{\mathrm{d}}{\mathrm{d}t}C_3 &= \frac{1}{i\hbar}\langle[S_x v_y - S_y v_x, H]\rangle && (66)\\
&= \frac{1}{i\hbar}\langle i\hbar\omega_c\left(1+a\right)S_y v_y - i\hbar\omega_c v_x S_x + i\hbar\omega_c\left(1+a\right)S_x v_x - i\hbar\omega_c S_y v_y\rangle && (67)\\
&= \Omega\langle S_y v_y + S_x v_x\rangle && (68)\\
&= \Omega C_2 && (69)\\
& && (70)
\end{aligned}$$

in total we have

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}C_1 &= 0 && (71)\\
\frac{\mathrm{d}}{\mathrm{d}t}C_2 &= -\Omega C_3 && (72)\\
\frac{\mathrm{d}}{\mathrm{d}t}C_3 &= \Omega C_2 && (73)
\end{aligned}$$

differentiating the equation for $C_3$

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}C_3 = \Omega\frac{\mathrm{d}}{\mathrm{d}t}C_2 = -\Omega^2 C_3 \tag{74}$$

whose solutions are a linear combination of sines and cosines. therefore

$$\begin{aligned}
C_1 &= A && (75)\\
C_2 &= D\sin\Omega t + E\cos\Omega t && (76)\\
C_3 &= E\sin\Omega t - D\cos\Omega t && (77)
\end{aligned}$$

since $C_2, C_3$ have opposite signs in the first order and depend on each other.

## 2.3 S dot v

$$\begin{aligned}
\langle\boldsymbol{S}\cdot\boldsymbol{v}\rangle &= \langle S_x v_x + S_y v_y + S_z v_z\rangle && (78)\\
&= C_1(t) + C_2(t) && (79)\\
&= C_1(0) + C_2(0)\cos\Omega t - C_3(0)\sin\Omega t && (80)
\end{aligned}$$

## 2.4 Experimental a

From the graph we can see the period is approximately 3.5 ţs therefore

$$a = \frac{2\pi}{\omega_C T} = \frac{2\pi m}{eBT} = \frac{2\pi \cdot 9.11 \times 10^{-31}}{1.6 \times 10^{-19} \cdot 9.4 \times 10^{-3} \cdot 3.5 \times 10^{-6}} =\sim 0.001 \tag{81}$$

## 2.5 comparison

The theoretical value is

$$a = 0.001195 \tag{82}$$

so the experimental value is very close.